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Research on the Operational Optimal Feasibility Domain of Compressed Carbon Dioxide Energy Storage Systems Based on Machine Learning

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Abstract

Compressed carbon dioxide energy storage (CCES), as an emerging compressed gas energy storage technology, exhibits significant potential for enhancing renewable energy accommodation and improving the flexibility of integrated energy systems, owing to its superior thermophysical properties, environmental friendliness, and capability for combined heat and power (CHP) supply. Based on an established thermodynamic model of a compressed carbon dioxide energy storage combined heat and power (CCES-CHP) system, this study proposed a machine learning methodology for the dynamic construction and optimization of its operational feasibility domain. Dimensionless factors γ_1 and γ_2 , along with ratio parameters α_1 and α_2 , are defined to quantify the coupling relationships among electrical, thermal, and mass flows. A dual state of charge (SOC) model incorporating both thermal and gas storage is developed to establish the system's constraint boundaries. Machine learning algorithms, including XGBoost and neural networks, are trained using thermodynamic simulation data to predict feasible operating regions and optimize system dispatch under varying load conditions. A case study on an integrated energy system (20 MW wind power, 30 MW conventional CHP, and 10 MW/50 MWh CCES-CHP) demonstrates that the proposed approach achieves 92.3% accuracy in predicting operational feasibility. The CCES-CHP system achieves electrical and total heating efficiencies of 66.7% and 21.4%, respectively, with a comprehensive energy utilization efficiency of 88.1%. By adjusting turbine inlet temperature, the system extends discharging duration from 5 to 8 hours under high electrical load scenarios, significantly improving operational flexibility. This methodology provides an effective tool for capacity planning and real-time dispatch of CCES-CHP systems.

Keywords

Compressed carbon dioxide energy storage; Combined heat and power; Operational feasibility domain; Machine learning; Integrated energy system